A minimax free-energy account of prioritized reactivation and systems consolidation in coupled predictive-coding networks

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Abstract

During sleep and quiet wakefulness, the brain spontaneously revisits neural activity patterns associated with previous experience, a phenomenon known as reactivation. Reactivation is thought to support systems consolidation, the process by which memory retrieval becomes progressively more dependent on neocortical and less dependent on hippocampal circuits. Evidence indicates that reactivation is not indiscriminate but preferentially targets particular memory traces.

Systems consolidation has motivated an extensive modeling literature addressing spontaneous reactivation, replay prioritization, and the hippocampo-neocortical redistribution of memory representations. In parallel, a growing literature has described hippocampal and neocortical circuits within the predictive-coding (PC) framework. Here, we connect these two lines of work by showing how coupled PC networks can generate autonomous intrinsic activity that redistributes stored memory content between neural systems.

We model the hippocampo-neocortical system as two coupled PC associative memory networks. The asymmetric coupling is inspired by the interaction modalities observed between hippocampal area CA1 and the prefrontal cortex, combining bottom-up recruitment with effective top-down suppression. This coupling establishes a closed-loop dialogue in which hippocampal activity trains the neocortical network, while neocortical feedback biases subsequent hippocampal activity toward content that remains poorly represented.

Under simplifying assumptions and after neural activity has settled, the coupled dynamics identify the mode in the hippocampal memory manifold with the largest remaining neocortical reconstruction error. A subsequent local neocortical plasticity event changes the synaptic weights so as to produce the largest immediate reduction in this error. Repeated reactivation and plasticity therefore greedily minimize the largest representational gap between the two systems.

In simulations with correlated memories, the architecture transferred memory content faster than random memory rehearsal. Together, these results provide a PC account on how the neocortical influence could guides hippocampal reactivation, prioritizing content that remains weakly represented.

Author summary

1 Introduction

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A large fraction of the brain's metabolic budget is devoted to intrinsic activity, that is, ongoing neuronal dynamics that cannot be traced back to recent sensory input or ongoing behavioural demands [1,2]. One of the better characterised events taking place during intrinsic activity is neural reactivation: the reinstatement, during sleep or quiet rest, of activity patterns associated with prior waking experience (REF). For a substantial fraction of these events, activity in the hippocampus and neocortex is coordinated, giving rise to what is termed the hippocampo-cortical dialogue [3,4].

This dialogue supports systems consolidation, the process by which memory retrieval becomes progressively less dependent on the hippocampus and more dependent on neocortical circuits [?, ?, 3, 5]. Systems consolidation is therefore often described as a redistribution or reorganization of memory representations across brain systems, or as a form of memory transfer whereby the neocortex progressively acquires memory content initially encoded in the hippocampus. (REF).

Extensive literature indicates that the reactivation events underlying systems consolidation preferentially target certain memory traces over others. Among over factors, traces that are recent or that correspond to novel experiences are prioritized for reactivation (REF). The precise neural mechanisms inducing this prioritization remain poorly understood (REF).

Reactivation and systems consolidation have motivated a broad range of computational models, from complementary learning systems [?] to autonomously reactivating attractor networks [?, ?, ?], generative replay [?, ?], and utility, context, or familiarity based replay scheduling [?, ?, ?, ?, ?]. These accounts differ in what drives reactivation, such as trace strength, recency, attractor competition, context, reward, or expected utility, and what information is transferred. Here, we focus on a single candidate mechanism, grounded in a specific communication motif between the hippocampus and neocortex.

The communication motif we consider has been most clearly characterized between the prefrontal cortex (PFC) and hippocampal area CA1, and combines two directional interactions: a bottom-up drive from CA1 to PFC and a top-down suppression from PFC to CA1. During NREM sleep, a brief synchronized burst within a CA1 neuronal assembly is associated with increased firing in functionally coupled PFC populations (REF). Conversely, when PFC populations reactivate independently, without a coincident CA1 burst, CA1 neurons that normally participate in coordinated CA1–PFC events are preferentially suppressed (REF). Together, these findings suggest an asymmetric, representation-selective dialogue: hippocampal reactivation recruits selected prefrontal populations, whereas prefrontal reactivation suppresses the corresponding hippocampal representation. This motif may extend beyond PFC–CA1: entorhinal projections also recruit local inhibitory circuits in CA1, providing an potential additional pathway for similar bi-directional interation (REF).(change place maybe)

Wang, Poe, and Zochowski gave this motif a functional interpretation: cortical memory representations regulate hippocampal reactivation according to their degree of transfer [6]. In their model, as a cortical representation forms during reactivation, it recruits hippocampal inhibitory interneurons that selectively suppress the corresponding hippocampal trace. As transfer proceeds, each trace withdraws its own reactivation, leaving replay to the content that still lacks cortical support.

We extend this account in three ways. First, we develop the functional role of feedback inhibition in organizing memory transfer. Second, we formulate the interacting hippocampal and neocortical memories within the predictive coding (PC) framework. Third, we show that, such dynamics effectively allows fast memory transfer targetting the largest representational gap between the two systems.

In the proposed model, neocortical feedback inhibition biases reactivation away from hippocampal content already supported cortically and toward content that remains poorly transfered. This dynamic directs the reactivation process toward the state belonging to the hippocampus memory manifold that maximize the neocortex settled free energy. Neocortical plasticity then reduce the corresponding free energy. The resulting architecture therefor gives rise to prioritized reactivation, transfering memory content as consolidation proceeds, and terminating reactivation once redistribution is complete.

More broadly, this work contributes to the study of fast and reliable offline information transfer between artificial neural networks.

We deliberately adopt a simplified model in which the hippocampus and neocortex are each represented by a single associative memory network composed of continuous-valued units rather than spiking neurons.

This simplification makes the model analytically tractable, allowing us to use matrix algebra to better understand the mechanism under study. The analysis is performed on a linear version of the network, and the results are then generalized numerically for nonlinear activation functions.

2 Models

In this section, we introduce the network architecture, beginning with the subnetworks local dynamics and following with their coupling modalities. We then present the network training and memory consolidation dynamics.

The model consists of two coupled associative-memory PC networks (PCNs), the teacher and student networks, corresponding respectively to the hippocampus and neocortex. Taken independently, each network performs retrieval by carrying out inference on a given cue, minimizing its local free energy through changes in neuronal activity. Learning is also induced by free energy minimization, this time through changes in its synaptic weights (REF). The full architecture adds an asymmetric coupling between the teacher and student networks.

Memory storage occurs in two successive phases. The teacher first stores externally acquired activity patterns, corresponding to fast memory encoding during awake activity; then, autonomous intrinsic activity of the coupled system allows transfer of the teacher memory content to the student network (Sec. 2.3).

2.1 A single predictive-coding associative memory

Tang et al. introduced the implicit covariance-learning PCN (implicit covPCN), which constitutes the basic building block of our architecture [7]. The implicit covPCN was developed as a more biologically plausible alternative to the explicit covPCN, whose learning rule requires non-local matrix operations and becomes numerically unstable on structured data [?, ?, ?, 7]. Here we present the linear version of the network which is the basis of the analytical study presented in this work.

Both the teacher T and the student S, corresponding respectively to the hippocampus and neocortex, are implicit covPCNs with dense recurrent connectivity and no self-connections (autapses).

We use $\alpha \in \{S, T\}$ to index either network. For activity $x_\alpha \in \mathbb{R}^d$ and recurrent weights W_α, we define the prediction-error operator and recurrent error

$$M_\alpha = I - W_\alpha, \qquad \varepsilon_\alpha = M_\alpha x_\alpha, \tag{1}$$

and the network free energy,

$$F_\alpha^{\text{rec}}(x_\alpha; W_\alpha) = \frac{\pi_\alpha}{2}\|\varepsilon_\alpha\|^2 = \frac{\pi_\alpha}{2}\|M_\alpha x_\alpha\|^2, \tag{2}$$

where $\pi_\alpha > 0$ is the precision of the prediction. Following Tang et al., we interpret the precision-weighted prediction-error objective as a simplified variational free-energy functional under fixed-precision Gaussian assumptions. We refer to it below simply as free energy.

F_α^{rec} accounts only for the network's internal prediction error generated by its recurrent connectivity. Network α lowers F_α^{rec} along two routes on two timescales: fast *inference* changes its activity, while slow *plasticity* changes its weights,

$$\tau_\alpha\, \dot{x}_\alpha = -\pi_\alpha\, M_\alpha^\top \varepsilon_\alpha, \qquad \dot{W}_\alpha = \eta\, \pi_\alpha\, \varepsilon_\alpha\, x_\alpha^\top, \qquad \tau_\alpha >> \eta, \tag{3}$$

with activity time constant τ_α and learning rate η. Weight updates and gradients with respect to the weights are understood to act only on the off-diagonal entries; self-connections remain fixed at zero.

The prediction may optionally pass through a pointwise nonlinearity f, giving $\varepsilon_\alpha = x_\alpha - W_\alpha f(x_\alpha)$. We take the linear case $f = \text{id}$ throughout the analysis and introduce a non linearity in Section **??**. In addition, as introduced in the next section, the teacher activity is normalized such that $\|x_T\| = 1$ after each integration step.

Each network is trained according to the plasticity dynamics in Eq. (3), either during interleaved clamping or through intrinsic activity during memory transfer (Sec. 2.3). In a fully trained network α, the stored content is the kernel of its error operator,

$$\mathcal{U}_\alpha = \ker M_\alpha = \{x \in \mathbb{R}^d :\ M_\alpha x = 0\}. \tag{4}$$

$\mathcal{U}_\alpha$ can therefor be interpreted as the network's memory manifold. Every stored pattern $m^{(p)}$ satisfies $M_\alpha m^{(p)} = 0$ and has zero recurrent free energy F_α^{rec} [7].

For the linear network, $\mathcal{U}_\alpha$ is a linear subspace. Any linear combination of stored memories is therefore a zero-energy fixed point. The network stores a flat memory manifold rather than a set of point attractors as in classical Hopfield networks (REF).

introduce a notation for the network memory manifold when we have normalization, later it will become easier to justify.

2.2 Coupling the teacher and the student

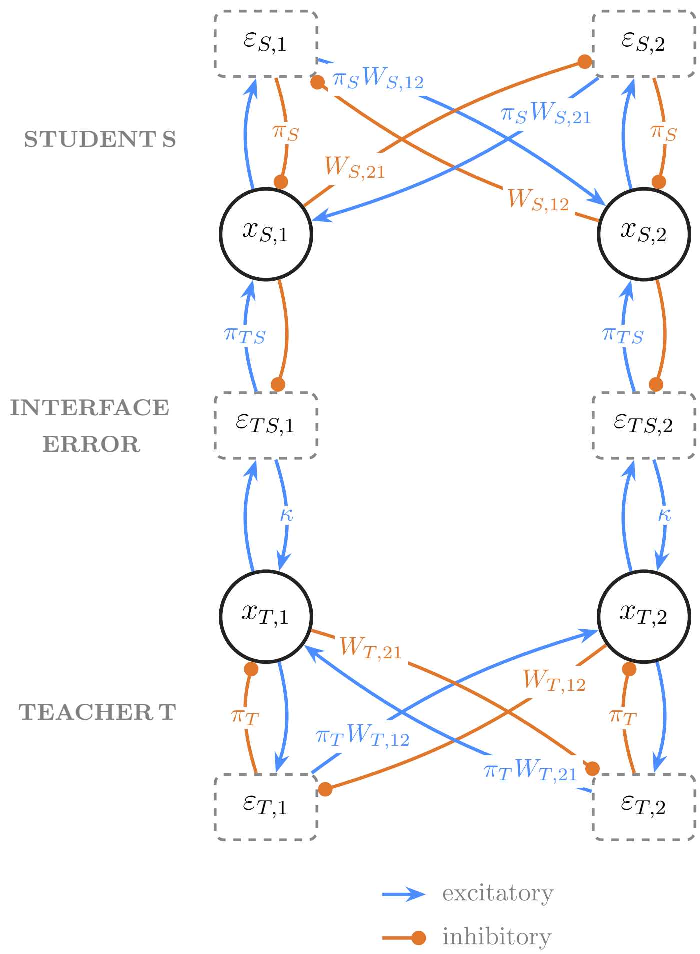

Fig 1. Model architecture (two units per population). Student S (top, cortical) above teacher T (bottom, hippocampal): value units x, recurrent errors $\varepsilon_S, \varepsilon_T$, and the one-to-one interface error $\varepsilon_{TS} = x_T - x_S$. Connections without labels have scalar value 1. During replay, the teacher-side interface gain κ makes the teacher ascend the student deficit rather than descending it as in classical PC networks.

Figure 1 shows the full architecture formed by coupling the student S and teacher T. The two networks interact through a one-to-one *interface error*,

$$\varepsilon_{TS} = x_T - x_S, \tag{5}$$

which measures the mismatch between the student state and the teacher state currently expressed. Without the interface influence, both the teacher T and the student S activities follow the inference dynamics in Eq. (3) and descends their own recurrent free energy F_T^{rec}.

For the teacher, the coupling adds a direct interface-error drive, giving

$$\tau_T \dot{x}_T = \underbrace{-\pi_T M_T^\top \varepsilon_T}_{-\nabla_{x_T} F_T^{\mathrm{rec}}} + \kappa \varepsilon_{TS} + \sigma_\xi \xi_T(t), \qquad \|x_T\| = 1, \tag{6}$$

where κ is the teacher-side coupling gain. During intrinsic activity, $\kappa > 0$, so the interface-error population excites the teacher. In conventional predictive coding, minimizing the interface error with respect to x_T would instead contribute $-\pi_{TS}\varepsilon_{TS}$ to the teacher dynamics. The teacher-side pathway therefore reverses the conventional sign. The term $\sigma_\xi \xi_T(t)$ provides low-amplitude stochastic input, with $\xi_T(t)$ denoting standard Gaussian white noise and σ_ξ setting its amplitude.

After each integration step, teacher activity is normalized to unit norm to prevent the system from producing unbounded activity. This operation idealizes fast population gain control: in biological circuits, divisive normalization mediated by recurrent and feedforward inhibition have been shown to perform this function [**?**].

The student side retains the conventional PC dynamics. Student activity minimizes its recurrent free energy and receives an excitatory influence from the interface-error population,

$$\tau_S \dot{x}_S = -\pi_S M_S^\top \varepsilon_S + \pi_{TS} \varepsilon_{TS}, \tag{7}$$

the student activity is not normalized the coupled dynamics.

Both contributions to Eq. (7) are gradient-descent terms on the student free energy which can therefore be expressed as

$$F_S(x_S, x_T; W_S) = \underbrace{\tfrac{\pi_S}{2} \|M_S x_S\|^2}_{F_S^{\mathrm{rec}}} + \underbrace{\tfrac{\pi_{TS}}{2} \|x_T - x_S\|^2}_{F_{TS}}, \qquad \tau_S \dot{x}_S = -\nabla_{x_S} F_S, \tag{8}$$

where $\pi_{TS} > 0$ is the precision of the interface prediction error. Activity and plasticity evolve concurrently in the simulations.

The overall influence between S and T deserve carefull reading. Although $\kappa > 0$ denotes an excitatory connection from the interface-error population to the teacher, the student enters $\varepsilon_{TS} = x_T - x_S$ with a negative sign, inhibiting the interface error. The combined pathway $x_S \to \varepsilon_{TS} \to x_T$ is thus inhibitory: increasing student activity reduces the excitatory error drive reaching the teacher. This effective top-down inhibition is the model counterpart of prefrontal suppression of hippocampal activity. In a conventional PC network, the interface error would instead inhibit the lower population, producing top down excitation through disinhibition.

2.3 Model dynamics

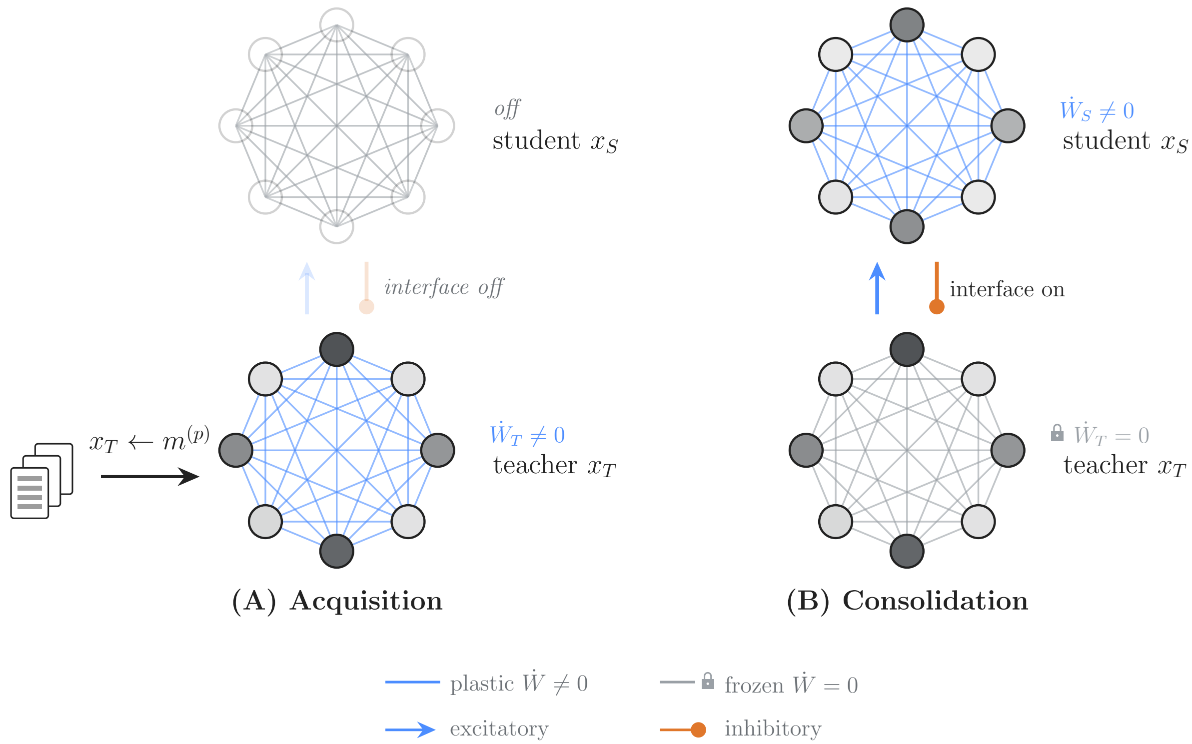

Fig 2. Initial memory aquisition and consolidation dynamics. (A) The teacher is trained on the memory corpus via interleaved rehearsal on the memory corpus. **(B)** Subsequent intrinsic dynamics transfer memory content from the teacher to the student.

Memory content is stored in the two subnetworks by the same local recurrent plasticity rule (Eq. 3), but under two different regimes generating neuronal activity. As shown in Fig. 2, the first phase establishes the teacher memory manifold; the second phase transfer this memory content to the student without presenting the student the original data.

During initial storage, Fig. 2**(A)**, teacher activity x_T is clamped to memory patterns $m^{(p)}$ in interleaved order while W_T follows Eq. 3. The student and the interface remain inactive. These updates reduce the teacher's recurrent prediction error on the presented patterns and establish its memory manifold $\mathcal{U}_T = \ker M_T$ (Methods). Interleaved training allows correlated memories to be stored while mitigating interference (REF). In contrast to our model, the hippocampus does not appear to require interleaved training to reliably store memory content during awake activity (REF). The orthogonalization of memory representations, particularly in the granule-cell layer of the dentate gyrus, is thought to support rapid learning of correlated memory traces (REF). This biological machinery is not part of our model and is therefore replaced by interleaved training.

During transfer (Fig. 2**(B)**), the external clamp is removed from the teacher and the interface is activated. The teacher and student activities follow Eqs. (6) and (7), while W_T remains fixed and only W_S follows the plasticity rule in Eq. 3 (Methods). The student is never clamped: intrinsic activity induces memory reactivation in the teacher network which subsequently influence the student through interface error. The student then display similar activity patterns and, finally, student recurrent plasticity allows the learning of these reactivated patterns.

For these dynamics to occure adequatly, the teacher network needs to stay on its memory manifold. To favor confinement to the memory manifold, we have consistently chosen throughout the paper

$$\kappa < \pi_T \sigma_{\min}^2, \tag{9}$$

where $\lambda_{\min}$ is the smallest nonzero eigenvalue of $M_T^\top M_T$, the strength of the teacher's weakest pull-back toward the manifold. Setting κ below it keeps the recurrent pull-back stronger than the interface drive in every off-manifold direction. Without this condition, for too large κ, the interface drive may push the teacher outside its memory manifold and toward random activation patterns, hindering good transfer.

3 Results

3.1 Reactivation and memory content redistribution

We first examined whether intrinsic activity in our model could reactivate and transfer stored memory content from the teacher to the student.

We stored three unit-normalized MNIST patterns in a linear 784-unit teacher network, froze the teacher's recurrent weights, and started the coupled dynamics with a plastic student whose recurrent weights were initialized to zero. During this intrinsic phase, neither network received external input. Low-amplitude Gaussian noise was applied to the teacher activity to seed the dynamics and avoid getting stuck in limit cycles. The results are shown in Fig. 3. Details on the parameter and shown metrics in Methods.

Fig. 3(A, bottom), (B, top), and (C) show that teacher activity is dominated by mixture states rather than by reactivation of individual patterns. This behavior follows from the linear activation function: the stored patterns span a flat memory manifold containing every signed linear combination of those patterns, including their antipodal states.

Fig. 3(D) and (E) provide direct evidence of memory-content transfer during intrinsic activity. In panel (D), $G_S(t) = U_T^\top M_S(t)^\top M_S(t) U_T$ measures the student's memory-reconstruction error within the teacher memory manifold, which is spanned by the orthonormal columns of U_T. Each plotted transfer-gap value is an eigenvalue of $G_S(t)$. Its associated eigenvector defines an orthogonal mode within the teacher memory manifold, generally a linear combination of several memory patterns, while the eigenvalue gives the squared reconstruction error for that mode.

The successive reduction of these modes shows that the student acquires the teacher's flat memory manifold one mode at a time, rather than by uniformly reducing the error across the entire manifold. Transfer rate slows once most of the manifold is stored by the student and remains incomplete over the displayed interval. Longer simulations further reduce the residual transfer gap, although at a much lower rate. At this late stage, the mismatch-driven selection signal is weak and the teacher increasingly relies on Gaussian-noise exploration to sample the remaining poorly represented modes. Despite this incomplete transfer, probing the reconstructions capacities of the student, as in Fig. 3 show that, after our transfer protocol, the student succesfully store the original patterns. Memory reconstruction capacities of the student are uneven at the partial checkpoint but allow recognizable reconstruction of all three memories later during transfer.

Fig. 3(B, bottom) and (E, middle and bottom) show how the signals driving learning and reactivation attenuate during memory transfer. The student plasticity rate decreases as the student–teacher mismatch contracts, revealing a self-limiting learning process: as the student acquires teacher-supported content, it removes the errors that generate further weight updates. In parallel, the teacher-side selection drive, generated by the mismatch-dependent top-down influence of the student on the teacher, also decreases. Three transient peaks accompany the successive reduction of the three eigenmodes in Fig. 3(D), consistent with the sequential acquisition of the three independent modes spanning the rank-three teacher memory manifold. Both signals remain nonzero at the end of the displayed interval, so this run approaches but does not reach exact self-extinction.

To test whether this mechanism extends beyond redistribution from a fixed teacher memory set, we allowed the teacher repertoire to grow during an otherwise continuous transfer run (Fig. 3). Each newly added memory introduced a new high-eigenvalue error mode, while the eigenvalues associated with memory modes already transfered to the student remained close to zero. These new gaps then declined, showing that the same mismatch-dependent dynamics can repeatedly redirect reactivation toward newly unsupported content. This protocol captures a feature of systems consolidation: ongoing experience continually adds new hippocampal memory content, which must then be integrated into neocortical circuits during subsequent periods of intrinsic activity (REF). In the model, each newly stored memory automatically recreates a hippocampo-cortical mismatch. Renewed prioritization therefore requires neither an explicit replay queue nor an externally assigned novelty tag. As cortical support for the new content develops, both the mismatch and the associated learning drive recede.

In summary, the coupled architecture transfer memory content from the teacher to the student during intrinsic activity. Transfer remains incomplete in the displayed run, yet the isolated-student probes show recognizable completion of all three stored patterns at the late checkpoint. Exact equality between the teacher and student memory manifolds is therefore not required for successful recall in this simple task. This result should nevertheless be interpreted as evidence of functional recall in a small associative-memory setup, rather than as a general demonstration of functional equivalence between the student and the teacher networks for more complex tasks. Tests involving classification of the reconstructed patterns, weaker cues, or more complex memory sets will be needed to determine when the residual transfer deficit becomes functionally limiting. In the Discussion, we consider possible mechanisms for improving redistribution exhaustiveness.

The observed dynamics are consistent with neocortex-dependent prioritization. The mismatch signal biases teacher activity toward teacher-supported modes that remain poorly represented by the student, and this bias weakens as those modes are learned. This provides a possible mechanistic account of the preferential reactivation of recently encoded hippocampal representations during post-learning sleep (REF), followed by reduced hippocampal involvement as memory retrieval becomes increasingly supported by neocortical circuits (REF). Nevertheless, the model does not treat recency itself as the priority signal. It predicts that reactivation depends more directly on the mismatch between hippocampal support and cortical representation.

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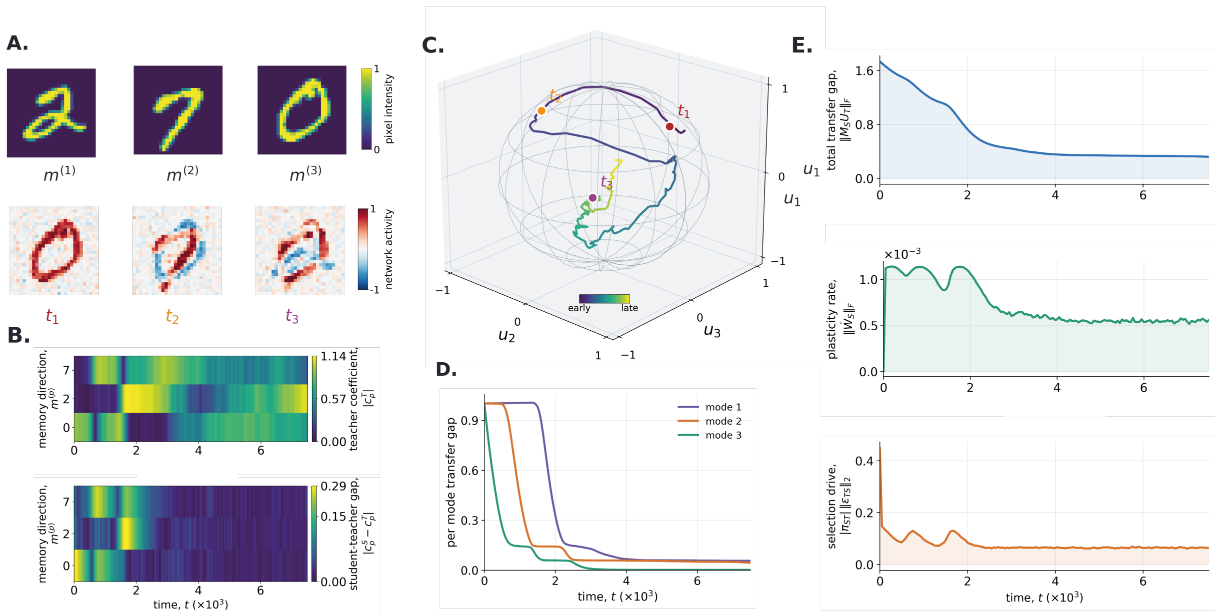

Fig 3. Spontaneous intrinsic activity progressively transfer teacher-supported memory content. The frozen teacher network was coupled to a student network initialized with zero activity and zero recurrent weights. Low-amplitude Gaussian noise seeded exploration of the teacher memory manifold. **A,** The three stored MNIST patterns (top; display order 2, 7, and 0) and teacher states at $t_1 = 250$, $t_2 = 1000$, and $t_3 = 7000$ (bottom). Although the stored patterns are positively signed, the linear network can express signed mixtures and antipodal states. **B,** (Top) Memory coefficient magnitude $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Method). Low values mean the teacher doesn't reactivates the memory while large values indicate strong reactivation. (Bottom) Student–teacher magnitude gaps $|c_p^S(t) - c_p^T(t)|$; a large gap indicates that the two networks express different magnitude along that pattern coordinate. **C,** Teacher activity projected onto an orthonormal basis U_T of the three-dimensional teacher memory space. The wireframe marks the unit sphere imposed by teacher normalization, and the colored markers identify the snapshots in panel A. Basis axes are orthogonal memory-space modes, not individual digits. **(D)** Per-mode transfer gaps, corresponding to the eigenvalues of $G_S(t) = U_T^\top M_S(t)^\top M_S(t) U_T$. High values identify modes of the teacher memory manifold that are poorly stored by the student, whereas low values indicate modes that are accurately stored. **E,** (Top) Total redistribution gap $\|M_S U_T\|_F$, which measures how much of the teacher memory space remains unsupported by the student. (Middle) Student plasticity rate $\|\dot{W}_S\|_F$, the magnitude of the current recurrent-weight update. (Bottom) Teacher-side selection drive $|\pi_{ST}|\|\varepsilon_{TS}\|_2$, the magnitude of the mismatch-dependent signal acting on the teacher.

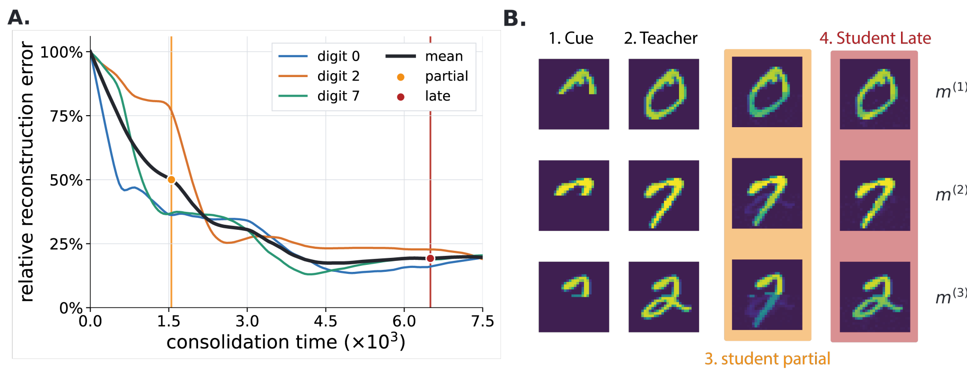

Fig 4. Transferred teacher memories become reconstructable by the isolated student. A, Relative recurrent reconstruction error for each stored MNIST pattern during consolidation. At each recorded time, the student weights were frozen, its state was set to the complete pattern $m^{(p)}$, and the error was evaluated as $r_p(t) = \|m^{(p)} - W_S(t)m^{(p)}\|_2 / \|m^{(p)}\|_2$. Thus, 100% corresponds to the initially empty student with $W_S = 0$, whereas zero denotes exact recurrent prediction of that pattern. The black curve is the mean over the three digits; orange and red markers identify the partial ($t = 1550$) and late ($t = 6500$) checkpoints used in panel B. **B,** Clamped recall by isolated, frozen networks. From left to right, each row shows the partial cue, the teacher reconstruction, and student reconstructions at the partial and late checkpoints. Half of the network units are clamped to a given memory pattern, the hidden half was initialized to zero, and the free units settled by minimizing the network free energy $\frac{\pi}{2}\|(I - W_\alpha)x_\alpha\|_2^2$, with no teacher–student coupling or ongoing plasticity. Student recall performances are uneven at the partial checkpoint, in agreement with the reconstruction error in panel A. For the late checkpoint, all reconstruction outputs visually approach the teacher completions despite a residual reconstruction error.

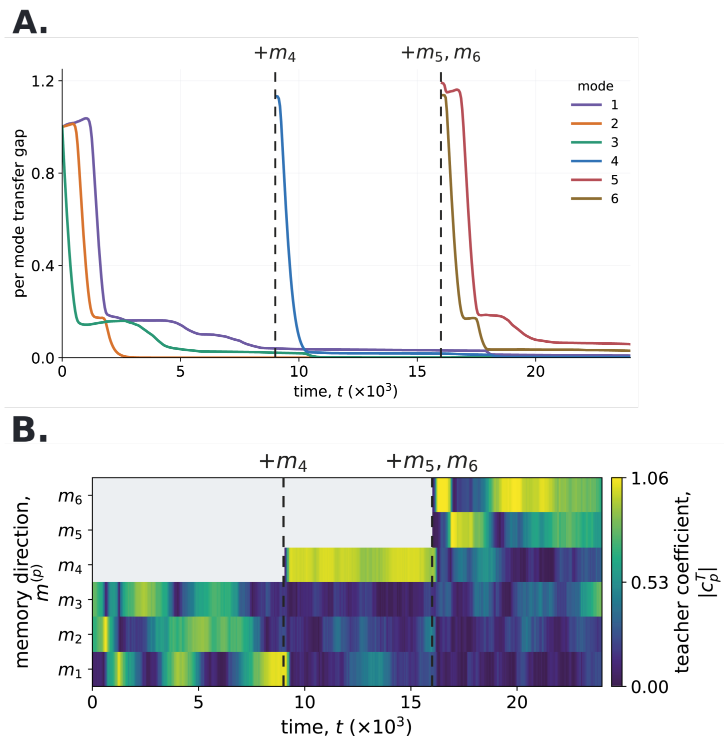

Fig 5. Staged memory acquisition renews reactivation and memory transfer. A 50-unit linear teacher–student network was followed through one continuous run with six random patterns. The teacher initially stored m_1, m_2, and m_3; it acquired m_4 at $t = 9000$ and acquired m_5 and m_6 together at $t = 16000$ through interleaved teacher training on the whole corpus. Coupled intrinsic dynamics resumed after each acquisition without resetting student activity or recurrent weights. Dashed lines mark the acquisition events. **A,** Per-mode transfer gaps during the simulation. Previously transferred modes keep a small error, whereas each newly supported mode introduces a large transfer gap that subsequently decreases. **B,** Memory coefficient magnitudes $|c_p^T(t)|$, which measure the contribution of each stored pattern $m^{(p)}$ to teacher activity $x_T(t)$ (Methods). Low values mean that the teacher does not reactivate the corresponding memory, whereas high values indicate strong reactivation. Newly added memories initially receive a stronger reactivation drive. As transfer progresses, this drive decreases.

3.2 Fast memory content transfer of structured data

Having established that the coupled network dynamics can transfer memory spaces, we next asked how efficiently this redistribution proceeds when the corpus contains unequally sized groups of correlated memories. We also ask how efficiently memory transfer is done during continuous incorporation of new patterns in the teacher stored corpus. We compared two protocols:

- **Intrinsic transfer (our method).** A trained, frozen teacher storing the memory space was coupled to an initially empty student. Teacher and student activity evolved under the coupled intrinsic dynamics, and the student updated its recurrent weights after each integration step. Optionally, memory patterns are added to the teacher memory corpus during transfer.
- **Stochastic rehearsal.** At each plasticity event (time step), one memory was sampled uniformly from the indexed corpus and clamped directly onto the student before the same local recurrent update was applied. Optionally, memory patterns are added to the indexed corpus during training.

Stochastic rehearsal emulates the transfer dynamics occuring from non-prioritized reactivation of memories, comparing it with our method therefore allows us to assess the impact of prioritized reactivation on the transfer dynamics.

The results for a correlated memory corpus, but without continuous incorporation of new patterns during learning, are shown in Fig. 6.

We considered a dataset of 26 memories divided in three correlated groups containing 18, 5, and 3 memories. Fig. 6(A) display the correlation structure of the memory set, memories within a group had a correlation of $\rho = 0.80$. The corpus construction, matched parameters, and implementation details are given in Materials and methods, Section 6.3.

We followed transfer at two levels. The named-memory residual used in the heatmaps asks whether the student can reconstruct each stored example. We also reported the normalized total transfer gap, $\|M_S U_T\|_F/\sqrt{P}$. It gives equal weight to every independent memory-space mode and measures how much of the complete teacher memory space remains unsupported by the student. Formal definitions of both measures are given in Materials and methods, Section 6.3.

The two protocols initially reduced the normalized total transfer gap at similar rates, but intrinsic transfer soon produced a faster decline than stochastic rehearsal (Fig. 6(B)).

Fig. 6(C) shows that intrinsic transfer reduced reconstruction errors at similar rates across all three groups, despite their unequal sizes, whereby stochastic rehearsal favor learning for the large clusters.

The difference in transfer rates reflects how each protocol selects the activity that drives plasticity. Stochastic rehearsal samples named memories uniformly. Because memories within each group are strongly correlated, these updates repeatedly reinforce their shared mode. This induces faster learning of over-represented groups. On the other hand, intrinsic transfer is not tied to named-memory frequency. As the student learns a teacher-supported mode, the mismatch for that mode contracts and its selection drive weakens, allowing activity to shift toward modes with larger remaining deficits. The result is consistent with deficit-dependent reactivation allocating plasticity according to reconstruction error across memory-space modes rather than the sample size of a given correlated memory-cluster.

Fig. 7 shows how within-group correlation affects the redistribution rate of both protocols while memory group sizes remain fixed. At low correlation, stochastic rehearsal was faster, whereas increasing correlation shifted the advantage toward intrinsic transfer.

Stochastic rehearsal can be more efficient when correlations are weak. Because the corpus acts as a lookup table, each plasticity event is applied directly to an exact memory. Intrinsic transfer instead updates the student while activity moves through the teacher memory space, and these intermediate states may produce partly redundant updates. Direct presentation therefore retains an advantage when uniform sampling already covers the memory subspace evenly, which happens when correlation fall.

Together, these results show that deficit-dependent intrinsic dynamics can redistribute highly correlated memory content with fewer plasticity events than stochastic rehearsal. This advantage grows with correlation, when uniform sampling increasingly produces overlapping updates. Such efficiency would matter for the brain as energy and opportunities for offline reactivation are limited (REF).

The next section presents a mathematical account of this qualitative behavior. We ask whether the coupled dynamics can be understood as optimizing an objective function derived from the PC formulation.

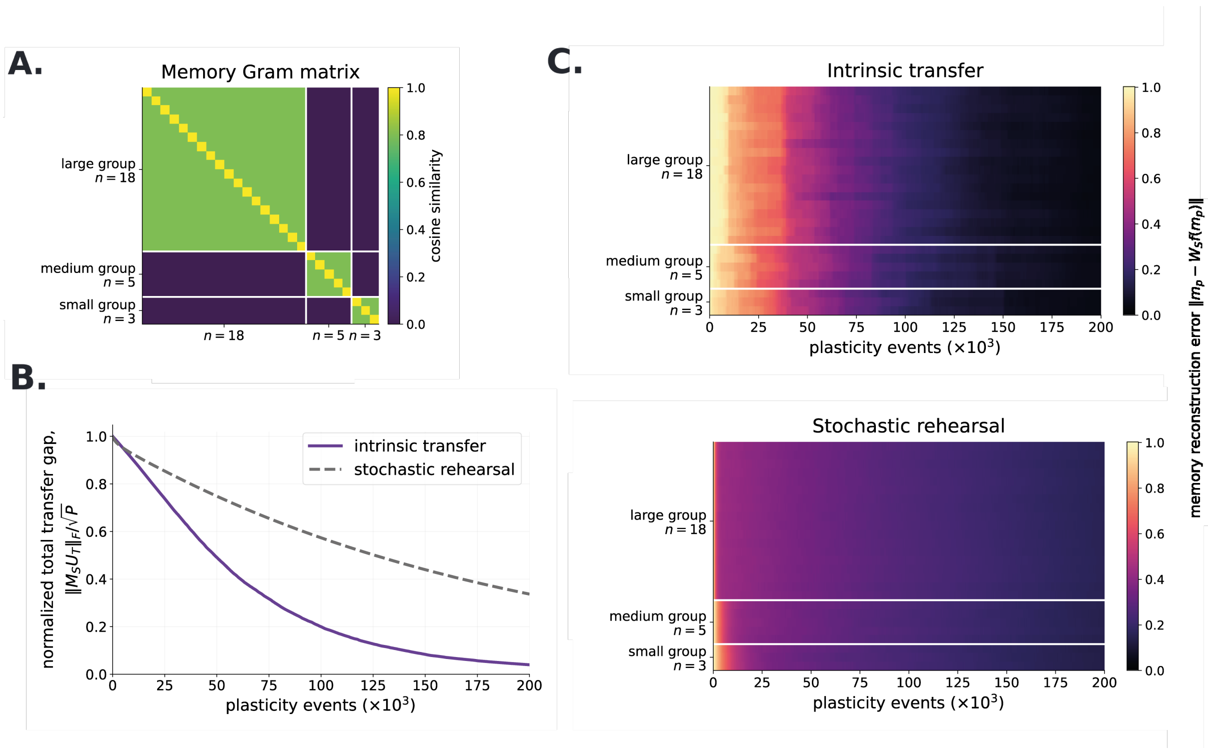

Fig 6. Intrinsic transfer accelerates redistribution across unequal groups of correlated memories. Both conditions used linear students initialized with $W_S = 0$, the same memory corpus, the same recurrent plasticity rule and rate, and 200,000 plasticity events. The corpus contained 26 memories divided into three mutually orthogonal groups of 18, 5, and 3 memories, with exact within-group cosine similarity $\rho = 0.80$. **A,** Memory Gram matrix. Each cell shows the cosine similarity between two memories. **B,** Normalized total transfer gap $\|M_S U_T\|_F/\sqrt{P}$ for intrinsic transfer (purple solid curve) and stochastic rehearsal (gray dashed curve). This measure gives equal weight to every independent memory-space mode, equals one for the initially empty student, and decreases toward zero as the student acquires the complete teacher memory space. After a short initial transient, intrinsic transfer reduced the gap faster than stochastic rehearsal. The rehearsal gap was still decreasing at the end of the protocol. **C,** Reconstruction errors for individual memories during intrinsic transfer (top) and stochastic rehearsal (bottom). Light colors indicate high reconstruction error and dark colors indicate accurate reconstruction. Intrinsic transfer reduced errors at similar rates across all three groups, whereas under stochastic rehearsal, errors remained higher for longer in the smaller groups. Metric definitions and full protocol details are provided in Materials and methods, Section 6.3.

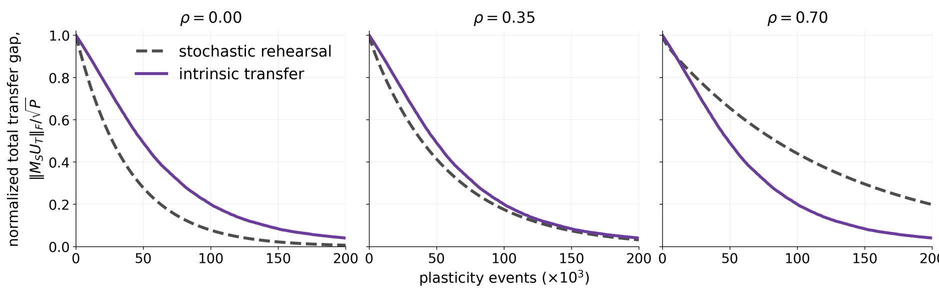

Fig 7. Increasing within-group correlation shifts the advantage from stochastic rehearsal to intrinsic transfer. The three conditions used the same group sizes (18, 5, and 3), student initialization, plasticity parameters, and number of plasticity events. Only the exact within-group cosine similarity changed, with $\rho \in \{0, 0.35, 0.70\}$. Each curve shows the normalized total transfer gap $\|M_S U_T\|_F/\sqrt{P}$, with lower values indicating more complete transfer. Gray dashed curves show stochastic rehearsal and purple curves show intrinsic transfer. At $\rho = 0$, stochastic rehearsal reduced the gap faster. At $\rho = 0.35$, the two protocols performed similarly. At $\rho = 0.70$, intrinsic transfer became faster. As correlation increased, stochastic rehearsal slowed because uniformly sampled memories produced increasingly overlapping updates, whereas the intrinsic-transfer curves changed comparatively little. All conditions used linear units. Full definitions and protocol details are provided in Materials and methods, Section 6.3.

3.3 Approximating minimization of the largest settled neocortical free energy

The preceding results show that coupled intrinsic activity can redistribute memory content from the teacher to the student. We now ask whether this process approximates minimization of the largest remaining student deficit over the teacher memory manifold.

We examine this question using a simplified protocol motivated by the separation of timescales

$$\tau_S \ll \tau_T \ll \frac{1}{\eta}. \tag{10}$$

In this regime, student activity relaxes much faster than teacher activity, while both relax much faster than the weights change.

Therefor, in this new protocol, the student settles while teacher activity stay fixed before each numerical integration step of the teacher dynamics. Both weight matrices remain fixed until teacher and student activity have converged, after which we apply one student-plasticity event. We call this complete sequence a *redistribution cycle*:

$$\underbrace{\text{student settles} \longrightarrow \text{teacher advances one step} \longrightarrow \text{student settles} \longrightarrow \cdots}_{\text{fixed weights, until activity convergence}} \longrightarrow \text{student plasticity.}$$

Once activity has converged, the final student state and its recurrent error determine one weight update according to Eq. (**??**). Teacher weights remain fixed throughout. The terminal teacher state initializes the next cycle, in which the student settles under its updated weights.

This protocol lets us test whether teacher activity approaches a state that maximizes the settled student free energy, and whether the subsequent local update follows gradient descent on this maximum.

As defined in Eq. (8), F_S combines the student's recurrent prediction error with the mismatch between teacher and student states. For fixed teacher activity x_T and student weights W_S, the linear student dynamics converge to the unique minimum of F_S with respect to x_S. We denote the free energy remaining after this relaxation by

$$q(x_T; W_S) = \min_{x_S} F_S(x_S, x_T; W_S). \tag{11}$$

Convergence to this minimum follows from the student's gradient dynamics and the strict convexity of F_S; it is therefore a property of the model (REF), rather than an additional hypothesis.

Hypothesis (1): During intrinsic activity, teacher activity settles near the state on its memory manifold at which the settled student free energy is maximal.

$$x_T^* \in \arg\max_{x_T \in \mathbb{S}_T} q(x_T; W_S). \tag{12}$$

and the corresponding student free energy,

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S), \tag{13}$$

with

$$\mathbb{S}_T = \{x_T \in \mathcal{U}_T : \|x_T\| = 1\}, \qquad \mathcal{U}_T = \ker M_T. \tag{14}$$

$\mathbb{S}_T$ correspond to all states in the teacher memory manifold, the unit-norm states in $\mathcal{U}_T = \ker M_T$.

The confinement of the teacher in its memory manifold is espected as

, the unit-norm states in $\mathcal{U}_T = \ker M_T$. Normalization is imposed, so the teacher state always has unit norm. Its confinement to $\mathcal{U}_T$, however, is only approximate: the recurrent dynamics damp off-manifold activity, but the interface drive can push it away.

To favor confinement to the memory manifold, we have consistently chosen throughout the paper

$$\kappa < \pi_T \sigma_{\min}^2, \tag{15}$$

where $\lambda_{\min}$ is the smallest nonzero eigenvalue of $M_T^\top M_T$, the strength of the teacher's weakest pull-back toward the manifold. Setting κ below it keeps the recurrent pull-back stronger than the interface drive in every off-manifold direction.

The largest settled student free energy on this reference set is

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} q(x_T; W_S) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_S(x_S, x_T; W_S). \tag{16}$$

Our hypothesis is that the teacher state attains this maximum,

$$x_T^* \in \arg\max_{x_T \in \mathbb{S}_T} q(x_T; W_S). \tag{17}$$

The reason to expect such selection is that, after student relaxation, the teacher-side interface drive is proportional to the gradient of q with respect to teacher activity, as shown below.

By definition,

$$q(x_T; W_S) \leq \mathcal{F}_{\max}$$

for every $x_T \in \mathbb{S}_T$. Therefor, a low maximum means that the student settles to low free energy for every unit state on the teacher memory manifold, whereas a large maximum means that at least one state remains poorly redistributed (or transfered). In the results commented bellow, we compare the full simulated teacher state after convergence with the nearest ideal state, so that both departures from the memory manifold and imperfect selection within it contribute to the measured distance.

Then we ask whether the activity selected during a redistribution cycle makes this same rule approximate descent on $\mathcal{F}_{\max}$. Where this maximum is differentiable, the prediction is

$$\dot{W}_S \approx -\eta \nabla_{W_S} \mathcal{F}_{\max}(W_S; W_T). \tag{18}$$

Here $\dot{W}_S$ is the local plasticity rate from Eq. (**??**), evaluated at the end of activity relaxation and applied only during the plasticity event. For a weight change of fixed Frobenius norm, the negative gradient gives the direction of largest first-order decrease in $\mathcal{F}_{\max}$. Student minimization is imposed by the relaxation protocol. Teacher maximization and descent on that maximum are the proposed effects of the teacher dynamics and local plasticity, which we test below.

This objective also describes the complete network free energy,

$$F_{\text{net}}(x_T, x_S; W_T, W_S) = F_T^{\text{rec}}(x_T; W_T) + F_S(x_S, x_T; W_S). \tag{19}$$

The teacher recurrent energy vanishes on $\mathbb{S}_T$, so

$$\mathcal{F}_{\max}(W_S; W_T) = \max_{x_T \in \mathbb{S}_T} \min_{x_S} F_{\text{net}}(x_T, x_S; W_T, W_S). \tag{20}$$

The largest settled student free energy is therefore also the largest settled free energy of the complete network over this manifold.

Testing the optimization performed during a cycle. The mathematical description gives two hypotheses for the redistribution cycle:

1. **Activity selection:** successive teacher steps, each evaluated after student relaxation, approach a teacher state close to an ideal state maximizing q over $\mathbb{S}_T$.

2. **Synaptic plasticity:** the local student update evaluated after activity relaxation approximates the ideal gradient-descent step on $\mathcal{F}_{\max}$.

Figure 8 tests these hypotheses by comparing observed activity states and weight-update directions with their ideal references, then follows the change in $\mathcal{F}_{\max}$ over successive redistribution cycles.

This construction is close in purpose to classical predictive coding, while differing in how the states on which free energy is minimized are obtained. PC networks ordinarily learn from activity elicited by current sensory input. During autonomous redistribution, no external input specifies the state to be learned. The network must organize both teacher-state selection and student learning. Teacher exploration may increase instantaneous network free energy, while the plasticity phase is expected to reduce its largest settled value over the teacher memory manifold.

The settled interface error is proportional to the gradient with respect to teacher activity. Student relaxation can be solved exactly in the linear model. The same calculation identifies how the resulting interface error drives teacher activity and motivates Hypothesis **1**. Define

$$S_S = M_S^\top M_S, \qquad H_S = \pi_{TS} I + \pi_S S_S, \qquad N_S = I - \pi_{TS} H_S^{-1} = \pi_S S_S H_S^{-1}. \tag{21}$$

Theorem 1 (Settled student response and interface-driven ascent)**.** *For the linear student with $\pi_S, \pi_{TS} > 0$, fixed teacher activity, and fixed weights, F_S has the unique global minimizer*

$$x_S^*(x_T) = \pi_{TS} H_S^{-1} x_T = (I - N_S) x_T. \tag{22}$$

The settled student free energy and its gradient with respect to teacher activity are

$$q(x_T; W_S) = \frac{\pi_{TS}}{2} x_T^\top N_S x_T, \qquad \nabla_{x_T} q = \pi_{TS} N_S x_T = \pi_{TS}(x_T - x_S^*). \tag{23}$$

Thus, when the student is at equilibrium and $\kappa > 0$, the teacher-side interface contribution is a positive multiple of this gradient:

$$\kappa \varepsilon_{TS}^* = \frac{\kappa}{\pi_{TS}} \nabla_{x_T} q. \tag{24}$$

Proof. For fixed x_T, the student gradient and Hessian are

$$\nabla_{x_S} F_S = H_S x_S - \pi_{TS} x_T, \qquad \nabla_{x_S}^2 F_S = H_S \succ 0.$$

The positive-definite Hessian ensures that student gradient flow at fixed teacher activity converges to $x_S^*(x_T)$. Solving the stationarity equation gives Eq. (22). Expanding the energy and using $H_S x_S^* = \pi_{TS} x_T$ gives

$$q = \frac{\pi_{TS}}{2} x_T^\top x_T - \frac{\pi_{TS}}{2} x_T^\top x_S^* = \frac{\pi_{TS}}{2} x_T^\top N_S x_T.$$

Since N_S is symmetric, differentiating it gives $\nabla_{x_T} q = \pi_{TS} N_S x_T$. Finally, $x_T - x_S^* = N_S x_T$, which identifies the gradient with the settled interface error. □

The theorem shows that, at each teacher step, the coupling to the settled student favors changes in teacher activity that increase the settled student free energy. Within the teacher memory manifold, this favors states that remain poorly represented by the student. This result alone does not establish that repeating these teacher steps approaches a state whose settled student free energy is close to the maximum over $\mathbb{S}_T$. This is the prediction of Hypothesis **1**, which we test numerically below.

Activity selection within a redistribution cycle. We first compared the teacher state reached during activity relaxation with the ideal state defined in Eq. (17). This reference was calculated directly from the fixed weights, independently of the simulated activity (Materials and methods, Section 6.4.1).

In a two-dimensional memory space, teacher trajectories from different initial states approached the two ideal states (Fig. 8A). We then tested selection across 20 independent networks with six-dimensional memory spaces. Students received increasing numbers of memory presentations and local plasticity updates. At each checkpoint, we tested the activity-relaxation phase of the redistribution cycle from ten random teacher states, keeping the weights fixed. The mean angle to the nearest ideal state was initially about 0.4°, but rose to about 1° at the final checkpoint, with larger variation across networks late in training (Fig. 8B). Two of the 6,000 bouts reached the relaxation-time limit; their endpoints remain included in the figure.

These results support Hypothesis **1**: the activity-relaxation phase approaches an ideal teacher state on average, although selection becomes less precise as the student learns. This trend is consistent with the reduction in selection drive already observed during redistribution (Fig. 3E). As the student better represents teacher content, the interface error decreases and the drive distinguishing the remaining poorly represented modes weakens. Activity can then change very slowly while still departing from the ideal mode. The late increase in angular error is therefore compatible with the same mechanism that initially directs reactivation toward the largest deficit.

Local plasticity within a redistribution cycle. We next tested whether the local update at the end of activity relaxation approximates gradient descent on the maximum. Using the same terminal student states, we compared the direction of $\dot{W}_S$ in Eq. (**??**) with $-\nabla_{W_S}\mathcal{F}_{\max}$. An angle of zero means that the two update directions agree. This tests the direction of the plasticity phase independently of its step size.

The mean update angle was initially about $0.5°$ and increased to about $3°$ at later checkpoints (Fig. 8C). Local plasticity therefore remained closely aligned with steepest descent on average, supporting Hypothesis **2**, while becoming less exact as activity selection departed from the ideal. Together, panels B and C show how the approximation changes with learning: the reduction in the student's deficit weakens teacher selection, and the resulting departure in activity carries into the direction of the local weight update.

Repeated redistribution cycles reduce the largest remaining deficit. The first two tests examine the activity and plasticity phases at independently prepared student checkpoints. We finally tested whether applying complete redistribution cycles successively reduces $\mathcal{F}_{\max}$. Both intrinsic transfer and random-state rehearsal began with a student that stored five of the six teacher memories, leaving one independent mode incompletely represented. Random-state rehearsal sampled unit states uniformly from the teacher memory manifold and used the same local plasticity rule and step size. This comparison asks whether intrinsic selection directs learning toward the remaining deficit when much of the memory space is already represented.

Intrinsic transfer reduced the maximum more rapidly over successive events (Fig. 8D). After 1,300 events, the median maximum settled free energy was about 1.5% of its initial value under intrinsic transfer, compared with 73% under random-state rehearsal. The decrease under intrinsic transfer slowed as the remaining deficit became smaller, consistent with the weakening of the drive for reactivation and plasticity in the preceding figures. Thus, approximate selection and local learning can still produce a sustained reduction in the largest deficit. The comparison measures progress per plasticity event, rather than per unit of relaxation time.

Complete redistribution would require

$$\mathcal{F}_{\max} = 0 \quad \Longleftrightarrow \quad \mathcal{U}_T \subseteq \ker M_S, \tag{25}$$

so that the student represents the entire teacher memory manifold. The maximum remains nonzero in this run, although it approaches a small fraction of its initial value. These results connect the effectiveness and the limits of redistribution to the same student-dependent drive: it prioritizes poorly represented content, then weakens as that content is learned. We now consider the implications of this mechanism for biological memory consolidation.

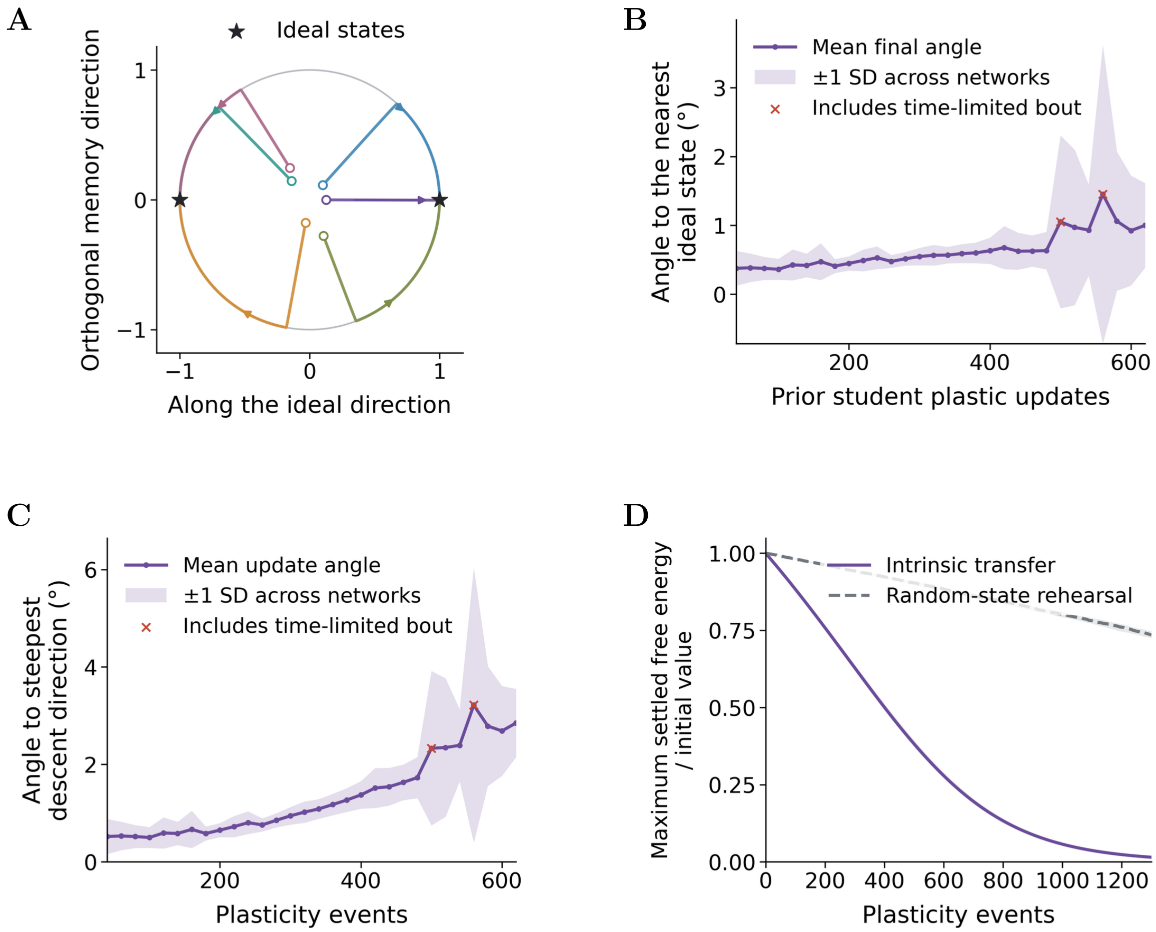

Fig 8. Coupled activity and local plasticity approximate minimization of the largest settled student free energy. Each redistribution cycle alternates student settling at fixed teacher activity and one teacher step, with weights held fixed until activity convergence, then applies one local student update. Ideal states and update directions were calculated independently from the weights. **A,** Six teacher trajectories projected onto a two-dimensional memory space without renormalization. The circle denotes unit memory states, stars the ideal states, open circles the starts, small dots the endpoints, and arrows the direction of time. **B,** Angle between the terminal teacher activity and the nearest ideal state. Students were prepared by prior named-memory presentations and plasticity updates, then tested with frozen weights by alternating student settling and teacher steps. The angle uses the full activity vector and treats both signs of an ideal mode as equivalent. **C,** Angle between the local student update and steepest descent of $\mathcal{F}_{\max}$ within zero-diagonal weights, using the same checkpoints and terminal states as B. Zero denotes identical directions; 90° denotes no first-order change in the maximum. The horizontal axis counts the prior training updates, not successive intrinsic cycles. In B and C, curves average ten initializations within each of 20 networks, then average across networks; shading is one standard deviation across network means. Red crosses mark checkpoints containing a bout that reached the relaxation-time limit; those endpoints remain included. **D,** Successive redistribution cycles from students storing five of six teacher memories. Intrinsic transfer repeats the redistribution cycle; random-state rehearsal uses the student's equilibrium response to a fresh uniform unit teacher-memory state. Both use $h = 0.005$. Curves show medians and bands the interquartile range across 20 networks, with each maximum divided by its initial value. Protocols are matched by event count. Reference calculations and protocols are given in Materials and methods, Sections 6.4.1–6.4.3.

4 Discussion

However, CA1–PFC representational coordination was weaker and less faithful to the waking experience during sleep than during awake ripples. The authors interpreted sleep reactivation as more integrative and less like precise reinstatement. CLEAR LIMITATION DUE TO THE FACT THAT WE USES SIMPLE ASSOCIATIVE MEMORY NETWORKS

4.1 Summary

[One paragraph: recipient deficit selects teacher content; selected content trains the recipient; learning removes the selecting signal. Prioritization, reduced reactivation of consolidated content, and self-extinction follow from the same objective.]

4.2 Relationship to other models

[Compare utility-based replay, context-driven replay, autonomous attractor consolidation, generative consolidation, recall-gated consolidation, and Go-CLS by their priority variable, autonomous selection mechanism, and what is transferred. The narrow novelty claim is a closed loop from recipient-deficit objective to source selection and recurrent recipient storage.]

4.3 Awake prefrontal memory control may be reused during sleep

[Lead with awake retrieval suppression and retrieval-induced forgetting. During wake, PFC control can suppress unwanted or competing hippocampal retrieval. Propose that related effective circuitry is reused offline: behavioral goals select during wake, whereas recipient competence selects during consolidation. Link to NREM PFC-associated suppression of CA1 activity and reactivation as evidence for a suppressive motif, not proof that PFC computes $\mathcal{D}_{\max}$.]

4.4 Experimental predictions, limitations, and future directions

[Predictions: cortical similarity suppresses matching hippocampal reactivation after controls for source strength, recency, salience, and reward; disrupting top-down suppression increases redundant reactivation and reduces efficiency; priority returns after cortical degradation. Limitations: linear subspaces; restricted ReLU extension; no ordered replay; idealized constraints and timescales; degeneracy; transformed codes; omitted source reliability, context, reward, emotion, salience, and neuromodulation. State explicitly that the analytic efficiency claim is local, that the exact $2/(1-\chi)$ factor assumes unconstrained weights and an already stored common component, and that faster complete transfer is supported numerically rather than by a global convergence or hitting-time theorem. Continual learning is a separate downstream project.]

5 Conclusion

[Two or three sentences. Close on the biological and mathematical message: cortical dependence can organize hippocampal reactivation; effective cortical suppression of the teacher follows from maximizing recipient deficit; and the discrepancy-driven prioritization signal vanishes at complete transfer.]

6 Materials and methods

6.1 Memory construction and network initialization

[Pattern generation, covariance-memory construction, dimensions, rank, correlation, recipient familiarity, seeds, and no-autapse constraint.]

6.2 Interleaved training of the teacher

6.3 Structured-memory transfer comparison

6.3.1 Controlled imbalanced corpus

The fast-transfer experiments used $P = 26$ unit-normalized memories arranged into three mutually orthogonal groups of sizes $n_g \in \{18, 5, 3\}$. The complete teacher memory space contained 26 independent modes. We assigned three neurons to each mode, giving $d = 78$ teacher and student units. A fixed permutation of neuron coordinates removed any visible block structure from the recurrent weights without changing the memory geometry.

Within group g, we specified the memory Gram matrix

$$G_g(\rho) = (1-\rho)I_{n_g} + \rho\,\mathbf{1}\mathbf{1}^\top, \qquad \mathcal{M}_g = B_g G_g(\rho)^{1/2}, \tag{26}$$

where the columns of B_g form an orthonormal basis for group g, the bases of different groups are mutually orthogonal, and $G_g(\rho)^{1/2}$ is the positive symmetric square root. Thus every memory has unit norm and every distinct pair within a group has cosine similarity ρ. For $0 \le \rho < 1$, each group has full rank n_g, and the complete memory span remains fixed as correlation changes. The main comparison used $\rho = 0.80$. The correlation sweep used $\rho \in \{0, 0.35, 0.70\}$ while keeping the group bases, sizes, rank, network parameters, and random seeds fixed. All simulations used linear units.

The Gram matrix in Fig. 6(A) is $G = \mathcal{M}^\top\mathcal{M}$, where $\mathcal{M}$ contains the 26 memories as columns. Because every memory has unit norm, G_{pq} is their cosine similarity. The three diagonal blocks show within-group redundancy, whereas the zero-valued off-block entries follow from orthogonality between groups.

6.3.2 Matched transfer protocols

We constructed the frozen teacher weights from the same 26-memory corpus in both conditions. Both students started with $W_S = 0$ and used linear units, plasticity rate $\eta = 0.04$, integration interval $\Delta t = 0.05$, and one projected recurrent-weight update per plasticity event. The diagonal of W_S was reset to zero after every update. Each run contained 200,000 events. The state time constants were $\tau_S = 0.1$ and $\tau_T = 1$, the gains were $\pi_T = 1$, $\pi_S = 0.5$, $\pi_{TS} = 1$, and $\pi_{ST} = -0.5$ in the implementation sign convention. Teacher noise had amplitude 0.05, and teacher activity was normalized around unit norm.

In the stochastic-rehearsal condition, an index was drawn uniformly from the 26 named memories at each event. The selected memory was clamped directly onto the student before applying its recurrent update. In the intrinsic-transfer condition, neither network was clamped. The teacher and student states advanced by one integration step under the coupled dynamics, after which the student received the same local recurrent update. Teacher weights remained fixed in both conditions. The coordinate permutation used seed 4, model construction used seed 12, the stochastic-rehearsal schedule used seed 23, and stochastic state evolution used seed 12,357.

Uniform sampling over named memories weights the expected random-presentation update by the empirical second moment

$$C_m = \frac{1}{P}\sum_{p=1}^{P} m_p m_p^\top. \tag{27}$$

The 18-memory group therefore contributes more updates than the 5- and 3-memory groups. Stronger within-group correlation concentrates these updates onto each group's shared mode, while the remaining contrast modes receive weaker updates.

6.3.3 Transfer measures

For each named memory, we measured the recurrent reconstruction residual

$$r_p(t) = \left\| m_p - W_S(t) f(m_p) \right\|_2. \tag{28}$$

The heatmaps in Fig. 6(C) show every $r_p(t)$ in group order. Under the linear activation used here, $f(m_p) = m_p$.

Let $U_T = [u_1, \ldots, u_P] \in \mathbb{R}^{d\times P}$ be an orthonormal basis of the teacher memory space, with $P = 26$, and let $M_S(t) = I - W_S(t)$. We used the normalized form of the total transfer gap introduced in Fig. 3,

$$\Delta_{\mathrm{tr}}(t) = \frac{\|M_S(t)U_T\|_F}{\sqrt{P}} = \left(\frac{1}{P}\sum_{j=1}^{P}\|M_S(t)u_j\|_2^2\right)^{1/2}. \tag{29}$$

This basis-independent measure gives equal weight to every independent memory-space mode. The gap equals one when $W_S = 0$, and it vanishes exactly when the student reconstructs the complete teacher memory space. Figures 6(B) and 7 report this normalized total transfer gap.

6.4 Reference states and numerical tests of redistribution

Figure 8 compares the states selected by coupled activity and the resulting local updates with their ideal counterparts. We used linear units and fixed teacher weights throughout. Student weights were held fixed during each activity-relaxation bout, so that selection and plasticity could be assessed separately.

6.4.1 Calculating an ideal teacher state

Let U_T have orthonormal columns spanning the teacher memory space $\ker M_T$. We obtained this basis from the right singular vectors of M_T with singular values below 10^{-9}. Any unit teacher-memory state can be written as $U_T a$, with $\|a\| = 1$. Restricting the settled-energy operator to this space gives

$$A_S = U_T^\top N_S U_T, \qquad q(U_T a; W_S) = \frac{\pi_{TS}}{2} a^\top A_S a. \tag{30}$$

The ideal mode is therefore a leading eigenvector v_1 of A_S, and

$$x_T^* = U_T v_1, \qquad \mathcal{F}_{\max} = \frac{\pi_{TS}}{2}\lambda_{\max}(A_S). \tag{31}$$

Both signs of x_T^* give the same energy. This reference depends only on the weights and does not guide the simulated activity. We recalculated it whenever student weights changed, using Cholesky solves to evaluate the equilibrium response. Since N_S is positive semidefinite and $\ker N_S = \ker M_S$, the maximum vanishes exactly when $M_S U_T = 0$, as stated in Eq. (25).

6.4.2 Activity and update angles

For panel B, let Q_* span the ideal teacher eigenspace. The nearest ideal state to the full unit teacher state x_T, and its angular distance in degrees, are

$$q_* = \frac{Q_* Q_*^\top x_T}{\|Q_*^\top x_T\|}, \qquad \theta_T = \frac{180}{\pi}\, 2\arcsin\!\left(\frac{\|x_T - q_*\|}{2}\right). \tag{32}$$

A zero projection gives 90°. Eigenvalues within $10^{-11} + 10^{-9}|\lambda_{\max}|$ of the maximum were treated as tied. The full activity vector was used, so the angle includes departures from the memory manifold. For the empty-student control, every unit memory state is ideal, and the distance tests approach to the manifold alone.

For the numerical calculations below, we write the existing plasticity rule as $G_{\mathrm{local}} = \dot{W}_S/\eta = \pi_S P_0[\varepsilon_S x_S^\top]$, where P_0 explicitly sets diagonal entries to zero. A discrete event is $\Delta W_S = hG_{\mathrm{local}}$, with $h = \eta\Delta t_{\mathrm{pl}}$. These are implementation expressions for the rule in Eq. (**??**).

For panel C, we calculated the exact student response $y_* = x_S^*(x_T^*)$ to an ideal teacher state. When the leading eigenvalue is simple, differentiating the settled maximum gives its steepest-descent direction within zero-diagonal weights,

$$G_* = -P_0\nabla_{W_S}\mathcal{F}_{\max} = \pi_S P_0\big[(M_S y_*)y_*^\top\big]. \tag{33}$$

The dependence through the minimizing student and maximizing teacher states contributes no derivative by stationarity. We compared G_* with G_{local} evaluated at the observed terminal student activity using the Frobenius angle

$$\theta_W = \frac{180}{\pi} \arccos\left(\frac{\langle G_*, G_{\mathrm{local}} \rangle_F}{\|G_*\|_F \|G_{\mathrm{local}}\|_F} \right), \qquad \langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij}. \tag{34}$$

The cosine was clipped to $[-1, 1]$ for roundoff, without taking its absolute value. An angle below 90° indicates a descent direction; above 90°, the update is uphill to first order. This comparison tests direction independently of update magnitude or step size. All sampled partial-student checkpoints had a simple leading eigenvalue and update norms above 10^{-12}, so the reference angle was defined.

6.4.3 Simulation and plasticity protocol

Networks and student preparation. We used 20 independent networks with 48 units per population and six linearly independent unit memories with pairwise cosine similarity 0.7. Reduced QR decomposition of a Gaussian matrix gave an orthonormal basis Q. Memories were the columns of $P = QL^\top$, where $LL^\top = 0.3I + 0.7\mathbf{1}\mathbf{1}^\top$. With $C = QQ^\top$, the teacher predictor was

$$(W_T)_{ij} = \frac{C_{ij}}{1 - C_{ii}} \quad (i \neq j), \qquad (W_T)_{ii} = 0. \tag{35}$$

This construction stores the memory space exactly and leaves student learning as the source of changing deficits.

For panels B and C, students began at $W_S = 0$. Each preparation event sampled a named memory uniformly, fixed the teacher at that memory, set the student to its equilibrium response, and applied the local update with $h = 0.25$. We tested frozen students after 40 to 620 updates in increments of 20. These checkpoints vary how well the student represents the teacher memories before intrinsic selection is tested. At each checkpoint, activity selection began from ten random full-sphere teacher states, with the student first settling to each teacher state. The same teacher starts were reused across checkpoints within each network. Panel A used a separate rank-two network, 160 preparation events, and six starts.

Activity relaxation and stopping. The redistribution cycle alternates student relaxation and teacher activity steps while holding both weight matrices fixed. For the current teacher state, student activity settles to

$$x_S^*(x_T) = \pi_{TS} H_S^{-1} x_T.$$

The teacher then advances using this response. With $S_T = M_T^\top M_T$, its norm-preserving flow is

$$v_T = \frac{-\pi_T S_T x_T + \kappa(x_T - x_S^*(x_T))}{\tau_T}, \qquad \dot{x}_T = v_T - x_T \frac{x_T^\top v_T}{x_T^\top x_T}. \tag{36}$$

After each teacher step, the student settles again at the new teacher state before another step is taken. This implements student equilibration within teacher relaxation, rather than integrating the two activity equations simultaneously. Teacher activity is not projected onto the memory manifold, and no noise or plasticity is applied during this alternation. A plasticity event uses the final settled student state only after teacher activity meets the stopping criterion.

Repeated redistribution and aggregation. For panel D, both protocols began with the same student weights, constructed by Eq. (35) from the span of the first five memories. Each protocol applied 1,300 updates with $h = 0.005$. Intrinsic transfer relaxed the coupled activity before each update and carried terminal activity into the next cycle. Only the first cycle began from a random teacher state; the student settled to the current teacher state at the start of every cycle. Random-state rehearsal drew a fresh uniform unit vector $a \in \mathbb{R}^6$, fixed $x_T = U_T a$, and used the exact student equilibrium for the same local update. This reference samples the memory space uniformly, so the comparison tests the benefit of directing learning toward the remaining deficit.

In B and C, we averaged angles over the ten starts within each network, then reported the mean and sample standard deviation across the 20 network means. All endpoints were included. In D, we recomputed $\mathcal{F}_{\max}$ after every update, divided by each network's initial maximum, and reported the median and interquartile range across networks. Event counts match plasticity opportunities, not elapsed relaxation time.

The workbench used NumPy `default_rng` with base seed 20260905 and independent network and protocol streams, recorded in the exported protocol. Numerical checks covered the settled-energy identity, a finite-difference weight gradient, zero diagonals, agreement with the repository dynamics, and selected trajectories at tenfold tighter solver tolerances and half the maximum step. A companion scalar check applied $h \in \{0.003, 0.001, 0.0003\}$ from identical pre-event states and compared the measured decrease per unit step with $\|G_{\text{local}}\|_F^2$. Relative errors omitted predictions below 10^{-12}; this separate check does not enter the update-angle measurement.

Supporting information

- **S1 Appendix.** Full proofs of Proposition **??**, Theorem **??**, Corollary **??**, and Corollary **??**; projected power iteration; envelope steps; antipodal maximizers; feasible projected descent; and the local correlated-memory rate comparison.
- **S1 Fig.** Ascent, zero-drive, and descent teacher-side coupling ablation.
- **S2 Fig.** Rank, dimension, correlation, and recipient-familiarity sweeps.
- **S3 Fig.** Timescale separation, noise, seed dependence, and teacher-manifold leakage.
- **S4 Fig.** Oracle implementation checks and random-baseline controls.
- **S5 Fig.** Rectified-network robustness and time-resolved individual-memory selectivity.

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